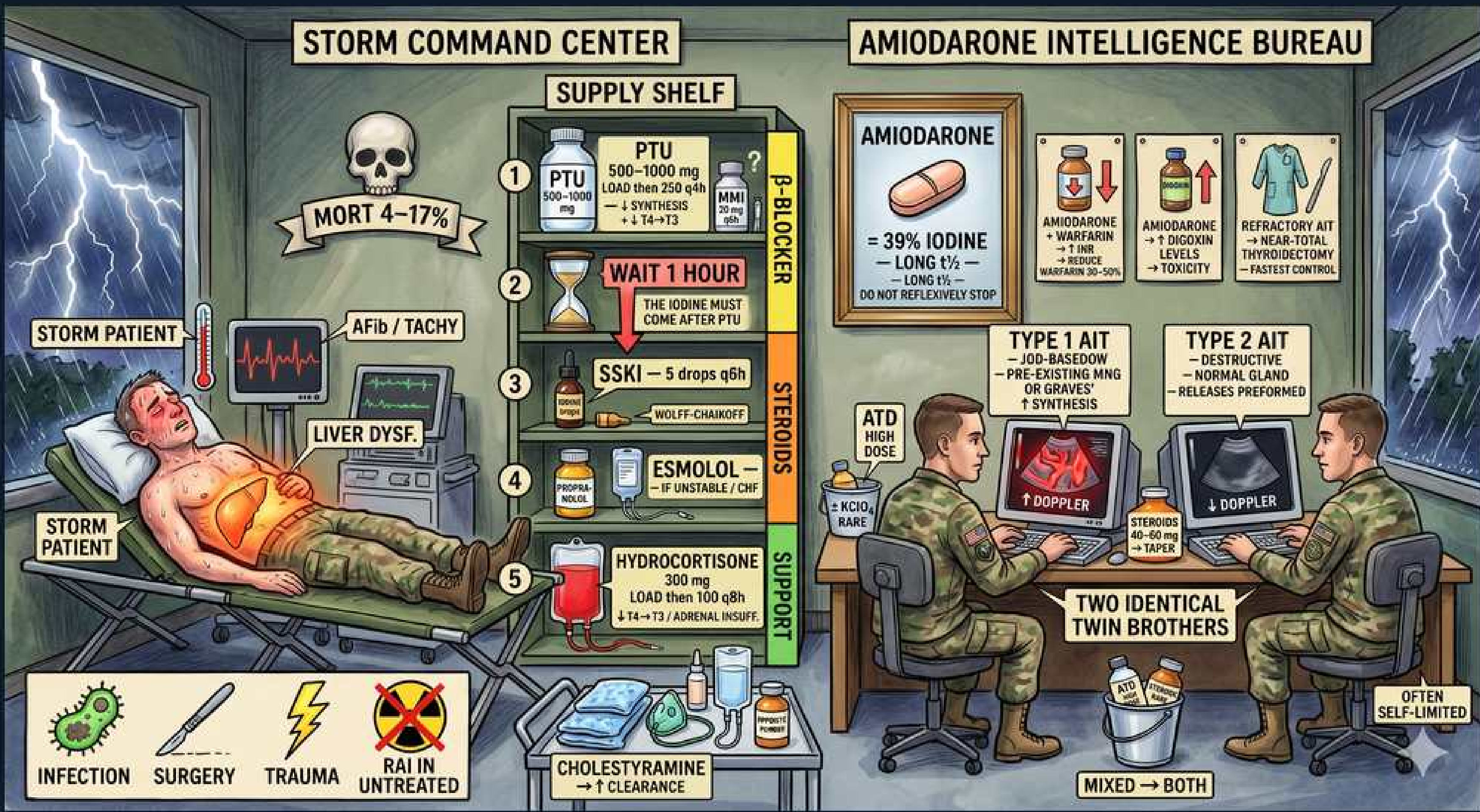
Thyroid — Thyroid Storm & Amiodarone (AIT)



1. High mortality

WHAT YOU SEE

Skull 'MORT 4–17%'

WHAT IT ENCODES

Thyroid storm is decompensated, life-threatening thyrotoxicosis with mortality ~10–30% (image: 4–17%) — an ICU emergency. It is a CLINICAL diagnosis (do not wait for labs, which look like ordinary hyperthyroidism); the Burch-Wartofsky Point Scale grades likelihood using fever, CNS dysfunction (agitation→coma), GI/hepatic features (nausea, jaundice), tachycardia/AF, and CHF (score ≥45 = highly suggestive).

BOARD TRAP

Waiting for confirmatory labs or a 'higher' free T4 than ordinary hyperthyroidism — storm is defined by organ dysfunction/decompensation, not by hormone level; treat empirically.

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2. AFib / tachycardia

WHAT YOU SEE

'AFib / TACHY' ECG icon

WHAT IT ENCODES

Marked sinus tachycardia (often >140) and atrial fibrillation with rapid ventricular response are cardinal features and a major driver of mortality, fueling high-output failure. Rate control with a beta-blocker is the first treatment step; thyrotoxic AF is also a stroke risk requiring anticoagulation consideration.

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3. Hepatic dysfunction

WHAT YOU SEE

'LIVER DYSF.' icon

WHAT IT ENCODES

Hepatic dysfunction (jaundice, transaminitis) is part of the Burch-Wartofsky picture and reflects hepatic congestion from heart failure plus the hypermetabolic state. Jaundice is a poor prognostic sign. It also complicates drug choice — PTU itself carries hepatotoxicity risk.

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4. 1. Beta-blocker first

WHAT YOU SEE

'β-BLOCKER' top of shelf

WHAT IT ENCODES

Step 1: a beta-blocker — propranolol (60–80 mg PO q4–6h or IV) is preferred because at high doses it also blocks peripheral T4→T3 conversion in addition to controlling adrenergic symptoms (tachycardia, tremor, anxiety, fever). Use cautiously/with esmolol in decompensated CHF.

BOARD TRAP

Withholding the beta-blocker out of fear of heart failure when the failure is rate-related — use short-acting IV esmolol so it can be titrated/stopped rather than giving nothing.

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5. 2. Thionamide (PTU)

WHAT YOU SEE

Shelf slot #1 holding a big PTU bottle '500–1000 then 250 q4h — ↓SYNTHESIS + ↓T4→T3' — the thionamide blocks new synthesis and must go in before iodine.

WHAT IT ENCODES

Step 2: a thionamide to block new hormone synthesis. PTU is preferred IN STORM (500–1000 mg load then 250 mg q4h) because it additionally inhibits peripheral T4→T3 conversion for faster effect. Methimazole is otherwise the everyday agent (longer half-life, less hepatotoxic) but lacks the conversion-blocking advantage acutely. The thionamide MUST be given before iodine.

BOARD TRAP

Giving iodine before the thionamide — iodine would then be used as substrate for new hormone synthesis (Jod-Basedow), worsening the storm.

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6. 3. Iodine — WAIT ≥1 hour

WHAT YOU SEE

An hourglass marked 'WAIT 1 HOUR — THE IODINE MUST COME AFTER PTU' on the supply shelf — the sand must run out before iodine is added, or iodine fuels new hormone.

WHAT IT ENCODES

Step 3: iodine (SSKI/Lugol's) but only ≥1 hour AFTER the thionamide. Once synthesis is blocked, the iodine load exploits the Wolff-Chaikoff effect to rapidly shut down hormone RELEASE. Given before the thionamide, the same iodine becomes raw material for new hormone (Jod-Basedow). Sequence: block synthesis → wait ≥1h → block release.

BOARD TRAP

Giving iodine first or simultaneously — without prior synthesis blockade it fuels new hormone production and accelerates the storm.

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7. SSKI Wolff-Chaikoff

WHAT YOU SEE

'SSKI — 5 drops q8h, WOLFF-CHAIKOFF'

WHAT IT ENCODES

Saturated solution of potassium iodide (SSKI, ~5 drops q6–8h) or Lugol's transiently inhibits both organification and hormone release (acute Wolff-Chaikoff effect). The effect is temporary (escape after ~1–2 weeks), so it is a bridge to definitive therapy. In iodine-allergic patients, lithium can substitute to block hormone release.

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8. 4. Esmolol / control

WHAT YOU SEE

'ESMOLOL — IV control / CHF'

WHAT IT ENCODES

IV esmolol is the preferred beta-blocker when tight titratability is needed — its ultra-short half-life lets you control rate in a patient with borderline cardiac function and stop instantly if heart failure worsens. It is the safer adrenergic-control option in thyrotoxic decompensated CHF.

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9. 5. Hydrocortisone (steroids)

WHAT YOU SEE

Shelf slot #5, a blood-red IV bag 'HYDROCORTISONE 300mg load then 100 q8h — ↓T4→T3 / ADRENAL INSUFF' on the SUPPORT band — steroids slow conversion and cover adrenal stress.

WHAT IT ENCODES

Glucocorticoids (hydrocortisone 100 mg IV q8h, or dexamethasone) serve two roles: they inhibit peripheral T4→T3 conversion and treat the relative adrenal insufficiency that accompanies the hypermetabolic stress of storm. They also help in autoimmune Graves'. Part of the standard 4-drug bundle: BB → thionamide → iodine → steroid.

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10. Cholestyramine adjunct

WHAT YOU SEE

A bottle on the lower table 'CHOLESTYRAMINE — ↑CLEARANCE' — it traps hormone in the gut to interrupt enterohepatic recycling and speed clearance.

WHAT IT ENCODES

Cholestyramine is an adjunct that binds thyroid hormone in the gut to interrupt enterohepatic recirculation and enhance fecal clearance, lowering circulating hormone faster in severe/refractory storm. Plasmapheresis is the last-resort option for refractory cases.

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11. Storm triggers

WHAT YOU SEE

'INFECTION, SURGERY, TRAUMA, RAI IN UNTREATED'

WHAT IT ENCODES

Storm is usually precipitated by an acute stressor in a hyperthyroid patient: infection/sepsis (most common), surgery (esp. thyroid surgery in an inadequately prepared patient), trauma, DKA, parturition, iodinated contrast, and radioactive iodine given to an UNTREATED hyperthyroid patient (RAI causes a transient hormone release). Pretreat with thionamides ± beta-blocker before surgery/RAI.

BOARD TRAP

Giving RAI to an untreated/poorly controlled hyperthyroid patient — the radiation-induced thyroiditis dumps stored hormone and can precipitate storm; render euthyroid first.

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12. Amiodarone = 37% iodine

WHAT YOU SEE

'AMIODARONE — 37% IODINE, LONG t½, WARFARIN/DIGOXIN interaction'

WHAT IT ENCODES

Amiodarone is ~37% iodine by weight (one tablet delivers ~50–100× the daily iodine requirement) and is highly lipophilic with a half-life of weeks to months, so thyroid effects persist long after stopping. It causes both hypo- and hyperthyroidism, and via CYP/transporter inhibition raises warfarin (↑INR — reduce dose) and digoxin levels. Check baseline TFTs before starting and monitor periodically.

BOARD TRAP

Assuming stopping amiodarone fixes amiodarone-induced thyrotoxicosis quickly — its enormous iodine pool and ultra-long half-life mean the thyroid effect lingers for months.

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13. AIT Type 1

WHAT YOU SEE

'TYPE 1 AIT — JOD-BASEDOW, PRE-EXISTING MNG/GRAVES, ↑SYNTHESIS'

WHAT IT ENCODES

Amiodarone-Induced Thyrotoxicosis Type 1 is iodine-induced EXCESS SYNTHESIS (Jod-Basedow) in an abnormal gland with pre-existing pathology (nodular goiter or latent Graves'). Color Doppler shows increased vascularity; RAIU is low-to-normal. Treat with high-dose thionamides (methimazole) ± potassium perchlorate to block iodine uptake.

BOARD TRAP

Treating Type 1 AIT with steroids alone — Type 1 is a synthesis problem requiring thionamides (± perchlorate); steroids are the answer for Type 2.

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14. AIT Type 2

WHAT YOU SEE

'TYPE 2 AIT — DESTRUCTIVE, NORMAL GLAND, releases preformed'

WHAT IT ENCODES

AIT Type 2 is a DESTRUCTIVE thyroiditis (drug-induced) in a previously NORMAL gland that releases preformed hormone, with low/absent Doppler flow and very low RAIU. Treat with GLUCOCORTICOIDS (prednisone), and it is often self-limited and may be followed by transient hypothyroidism — like other destructive thyroiditides.

BOARD TRAP

Giving thionamides as primary therapy for Type 2 — there is no excess synthesis to block; the treatment is glucocorticoids.

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15. Doppler distinguishes AIT

WHAT YOU SEE

Two identical twin-brother techs at side-by-side color-Doppler screens (one ↑flow, one ↓flow) — vascularity on Doppler, not RAIU, tells Type 1 (hyperfunctioning) from Type 2 (destructive).

WHAT IT ENCODES

Color-flow Doppler ultrasound is the key bedside test to separate the two: INCREASED vascularity = Type 1 (hyperfunctioning gland), absent/decreased vascularity = Type 2 (destruction). RAIU is usually low in both (because of iodine saturation), so Doppler — plus gland morphology and IL-6 — guides therapy.

BOARD TRAP

Relying on RAIU to differentiate AIT — uptake is low in both types because of the iodine load; Doppler vascularity is the discriminator.

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16. Mixed AIT → both

WHAT YOU SEE

A tag 'MIXED → BOTH' between the twin Doppler stations — overlapping Type 1 + Type 2 features, so treat with thionamide PLUS steroid together.

WHAT IT ENCODES

Mixed/indeterminate AIT has features of both mechanisms and is treated with combined thionamide PLUS glucocorticoid therapy. Whether to continue or stop amiodarone is individualized with cardiology — the drug's long half-life means stopping gives little short-term benefit, and definitive treatment (thyroidectomy) may be needed for refractory cases.
